

# **Mechanical Design and Analytical Verification of a Manual ACME Screw Scissor Jack**

Independent Machine Design Project  
Spring 2026

Alihan Acuner  
Department of Mechanical Engineering  
Karadeniz Technical University  
Trabzon, Türkiye

## Abstract

This project presents the mechanical design and analytical verification of a manually operated ACME screw scissor jack rated for a vertical load of 3307 lbf, corresponding approximately to a 1500 kg passenger vehicle application. The design was developed in Siemens NX using the IPS unit system. The study includes linkage kinematics, arm-force calculations, pin and bearing checks, arm buckling, ACME screw sizing, self-locking and torque calculations, threaded nut design, trunnion revision, axial screw support, handle sizing, and base and saddle checks. A design factor of 1.5 was generally applied through allowable stresses. Several local regions were revised after analytical checks, including the arm-end ligament, lower pivot region, and nut-block trunnion assembly. The resulting design satisfies the principal nominal strength and stability requirements, while local contact and stress-concentration regions remain subject to detailed finite-element analysis and proof-load testing.

## 1 Introduction

A scissor jack is a compact mechanical lifting device commonly used to raise passenger vehicles for tasks such as wheel replacement and tire inspection. Although the mechanism is visually simple, its design combines linkage kinematics, threaded power transmission, pin-joint loading, column stability, bearing pressure, and manufacturability.

The objective of this project is to design and analytically verify a manually operated ACME screw scissor jack rated for a vertical load of 3307 lbf, corresponding approximately to a 1500 kg passenger vehicle application. The design was developed in Siemens NX using the IPS unit system.

A representative commercial scissor jack is shown in Figure 1 as a visual reference for the initial concept.



Figure 1: Representative commercial scissor jack used as a visual reference for the concept study.

The final mechanism uses one moving threaded trunnion nut and one non-threaded support-side

block. The power screw rotates but is restrained against axial translation at the support-side block. Rotation of the screw causes the threaded nut to move horizontally, changing the scissor-arm angle and raising or lowering the upper saddle.

The report covers the working principle, kinematic layout, analytical load calculations, sizing of the arms, pins, screw, threaded nut, trunnion assembly, axial screw support, manual handle, base, and saddle. Critical regions requiring detailed finite-element and experimental verification are identified at the end of the report.

## 2 Design Objectives and Requirements

The principal design objectives were to obtain a compact manually operated jack, maintain self-locking behavior, use a formed U-channel arm geometry, and verify the major components analytically before finalizing the CAD model.

The final design requirements and selected dimensions are summarized in Table 1.

Table 1: Final design requirements and principal selections

| <b>Parameter</b>         | <b>Imperial Value</b>                 | <b>Metric Equivalent</b> |
|--------------------------|---------------------------------------|--------------------------|
| Rated vertical load      | 3307 lbf                              | 14.7 kN                  |
| Design factor            | 1.5                                   | —                        |
| Minimum external height  | 4.72 in                               | 120 mm                   |
| Maximum external height  | 15.0 in                               | 381 mm                   |
| Arm length, pin-to-pin   | 10.4857 in                            | 266.3 mm                 |
| Minimum arm angle        | 10.5 deg                              | —                        |
| Maximum arm angle        | 41.1 deg                              | —                        |
| Arm section              | $1.875 \times 0.800 \times 0.1875$ in | —                        |
| Main pivot pin diameter  | 0.500 in                              | 12.7 mm                  |
| Power screw              | 3/4–6 ACME, RH, single-start          | —                        |
| Arm and bracket material | HSLA Grade 80 steel                   | —                        |
| Screw and high-load pins | AISI 4140 HT steel                    | —                        |
| Threaded nut material    | C95400 aluminum bronze                | —                        |
| CAD software             | Siemens NX                            | —                        |

The design factor is applied through allowable stresses rather than by multiplying the rated load. Therefore, the rated load of 3307 lbf is used directly in the equilibrium equations, while allowable stresses are obtained by dividing material yield strengths by 1.5.

The external height differs from the internal pivot-center height because of the offsets introduced by the base, brackets, and upper saddle. These offsets are included in the completed CAD assembly.

### 3 Working Principle of the Scissor Jack

The jack is actuated by a right-hand, single-start 3/4-6 ACME power screw. One side of the mechanism contains a moving threaded trunnion nut. The opposite side contains a non-threaded support block that allows the screw to rotate while preventing axial translation.

When the user rotates the detachable crank handle, the threaded nut moves along the screw axis. This changes the horizontal distance between the side pivots and causes the scissor arms to rotate. As the side pivots move toward each other, the arm angle increases and the upper saddle rises. Reversing the screw rotation lowers the saddle.

The load path is

vehicle  $\rightarrow$  upper saddle  $\rightarrow$  upper pivots  $\rightarrow$  scissor arms  $\rightarrow$  side blocks and power screw  $\rightarrow$  lower pivots  $\rightarrow$  base  $\rightarrow$   
(1)

The smallest arm angle is the critical operating condition because it produces the largest axial forces in the arms and screw. The ACME thread was selected because it is more suitable for power transmission than a conventional V-thread and can provide self-locking behavior at the selected lead angle and friction coefficient.

### 4 Kinematic Layout and Geometry

The arms are treated as rigid two-force members connected by ideal pin joints. Each arm has a pin-center length  $L$  and forms an angle  $\theta$  with the horizontal. The internal vertical distance between the upper and lower bracket pivots is denoted by  $h$ , while the horizontal distance between the two side pivots is denoted by  $x$ .

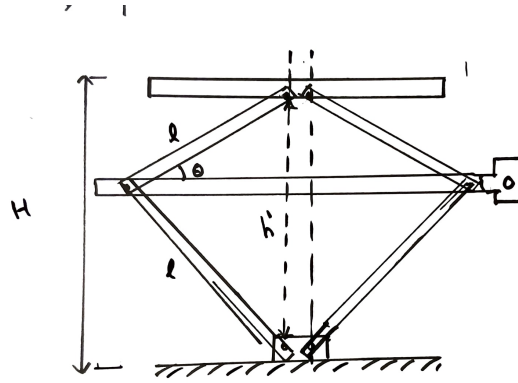


Figure 2: Simplified kinematic representation of the scissor mechanism.

For the symmetric linkage,

$$h = 2L \sin \theta \quad (2)$$

and

$$x = 2L \cos \theta. \quad (3)$$

The final pin-center arm length is

$$L = 10.4857 \text{ in.} \quad (4)$$

At the minimum operating angle,

$$\theta_{\min} = 10.5^\circ, \quad (5)$$

the internal pivot-center height is

$$h_{\min} = 2(10.4857) \sin 10.5^\circ \approx 3.82 \text{ in.} \quad (6)$$

At the maximum operating angle,

$$\theta_{\max} = 41.1^\circ, \quad (7)$$

the internal pivot-center height is

$$h_{\max} = 2(10.4857) \sin 41.1^\circ \approx 13.79 \text{ in.} \quad (8)$$

The corresponding side-pivot distances are

$$x_{\max} = 2(10.4857) \cos 10.5^\circ \approx 20.62 \text{ in.} \quad (9)$$

$$x_{\min} = 2(10.4857) \cos 41.1^\circ \approx 15.82 \text{ in.} \quad (10)$$

Thus, the required relative side-pivot movement is approximately

$$\Delta x = x_{\max} - x_{\min} \approx 4.80 \text{ in.} \quad (11)$$

The external jack height includes the structural offsets of the base, lower bracket, upper bracket, and saddle and is verified from the final CAD assembly.

## 5 Load Analysis

The vehicle load is assumed to act vertically and centrally on the upper saddle. The arms are modeled as two-force members, and pin friction is neglected in the equilibrium calculation.

The rated vertical load is

$$W = 3307 \text{ lbf.} \quad (12)$$

At an arm angle  $\theta$ , vertical equilibrium gives

$$F_{\text{arm}} = \frac{W}{2 \sin \theta}. \quad (13)$$

At the minimum angle,

$$F_{\text{arm,max}} = \frac{3307}{2 \sin 10.5^\circ} \approx 9070 \text{ lbf.} \quad (14)$$

The horizontal components of the upper and lower arms acting on the threaded side block add together. Therefore, the maximum axial screw force is

$$F_s = 2F_{\text{arm}} \cos \theta = W \cot \theta. \quad (15)$$

At the minimum operating angle,

$$F_{s,\text{max}} = 3307 \cot 10.5^\circ \approx 17.8 \text{ kip.} \quad (16)$$

The minimum-height configuration therefore governs the design of the arms, pins, screw, threaded nut, trunnion assembly, and axial support.

## 6 Pin and Bearing Stress Calculations

The upper and lower arm-to-bracket joints were designed using a 0.500 in diameter clevis pin. The maximum load acting on one scissor arm occurs at the minimum operating angle of  $\theta_{\text{min}} = 10.5^\circ$ .

The axial force carried by one arm is

$$F_{\text{arm}} = \frac{W}{2 \sin \theta_{\text{min}}} \quad (17)$$

where the rated vehicle load is

$$W = 3307 \text{ lbf.} \quad (18)$$

Thus,

$$F_{\text{arm}} = \frac{3307}{2 \sin 10.5^\circ} \approx 9070 \text{ lbf.} \quad (19)$$

The clevis pin was assumed to be manufactured from heat-treated AISI 4140 steel with an approximate yield strength of

$$S_y = 155 \text{ ksi.} \quad (20)$$

A design factor of

$$n = 1.5 \quad (21)$$

was applied through the allowable stress.

### 6.1 Double-Shear Check of the Pin

Since the arm is positioned between two bracket cheeks, the pin is loaded in double shear. The cross-sectional area of one shear plane is

$$A_p = \frac{\pi d_p^2}{4} \quad (22)$$

where

$$d_p = 0.500 \text{ in.} \quad (23)$$

Therefore,

$$A_p = \frac{\pi(0.500)^2}{4} = 0.1963 \text{ in}^2. \quad (24)$$

The average double-shear stress in the pin is

$$\tau_p = \frac{F_{\text{arm}}}{2A_p}. \quad (25)$$

Substituting the numerical values gives

$$\tau_p = \frac{9070}{2(0.1963)} \approx 23.1 \text{ ksi.} \quad (26)$$



Using the von Mises shear-yield approximation, the allowable shear stress is

$$\tau_{\text{allow}} = \frac{S_y}{\sqrt{3} n}. \quad (27)$$

Hence,

$$\tau_{\text{allow}} = \frac{155}{\sqrt{3}(1.5)} \approx 59.7 \text{ ksi}. \quad (28)$$

Since

$$\tau_p = 23.1 \text{ ksi} < \tau_{\text{allow}} = 59.7 \text{ ksi}, \quad (29)$$

the selected pin is satisfactory with respect to double shear.

## 6.2 Bracket Bearing Stress

Each bracket cheek carries approximately one-half of the arm force:

$$F_{\text{cheek}} = \frac{F_{\text{arm}}}{2}. \quad (30)$$

Therefore,

$$F_{\text{cheek}} = \frac{9070}{2} = 4535 \text{ lbf}. \quad (31)$$

The bracket-cheek thickness is

$$t_b = 0.1875 \text{ in}. \quad (32)$$

The projected bearing stress in each bracket cheek is

$$\sigma_{\text{br}} = \frac{F_{\text{cheek}}}{d_p t_b}. \quad (33)$$

Thus,

$$\sigma_{\text{br}} = \frac{4535}{(0.500)(0.1875)} \approx 48.4 \text{ ksi.} \quad (34)$$

The bracket was assumed to be manufactured from HSLA Grade 80 steel with a yield strength of

$$S_{y,\text{br}} = 80 \text{ ksi.} \quad (35)$$

The corresponding allowable normal stress is

$$\sigma_{\text{allow,br}} = \frac{S_{y,\text{br}}}{n}. \quad (36)$$

Therefore,

$$\sigma_{\text{allow,br}} = \frac{80}{1.5} = 53.3 \text{ ksi.} \quad (37)$$

Since

$$\sigma_{\text{br}} = 48.4 \text{ ksi} < \sigma_{\text{allow,br}} = 53.3 \text{ ksi}, \quad (38)$$

the bracket satisfies the selected design factor in bearing.

### 6.3 Pin Bending Check

The clear distance between the two bracket cheeks is

$$L_c = 1.50 \text{ in,} \quad (39)$$

while the central arm-eye thickness is

$$t_{\text{eye}} = 0.375 \text{ in.} \quad (40)$$

To reduce the unsupported pin length, two inward-projecting support sleeves were introduced. Each sleeve extends approximately

$$L_s = 0.525 \text{ in} \quad (41)$$

into the bracket opening.

The resulting effective central span is

$$L_{\text{eff}} = L_c - 2L_s. \quad (42)$$

Hence,

$$L_{\text{eff}} = 1.50 - 2(0.525) = 0.450 \text{ in.} \quad (43)$$

The total running clearance around the arm eye is therefore

$$c_{\text{total}} = L_{\text{eff}} - t_{\text{eye}} \quad (44)$$

and

$$c_{\text{total}} = 0.450 - 0.375 = 0.075 \text{ in.} \quad (45)$$

This corresponds to approximately

$$c_{\text{side}} = 0.0375 \text{ in} \quad (46)$$

on each side.

For a conservative preliminary assessment, the pin was idealized as a simply supported beam subjected to a concentrated load at midspan. The maximum bending moment is

$$M_{\text{max}} = \frac{F_{\text{arm}} L_{\text{eff}}}{4}. \quad (47)$$

Therefore,

$$M_{\max} = \frac{9070(0.450)}{4} \approx 1020 \text{ lbf} \cdot \text{in.} \quad (48)$$

For a solid circular pin, the maximum bending stress is

$$\sigma_{b,p} = \frac{32M_{\max}}{\pi d_p^3}. \quad (49)$$

Substituting the numerical values gives

$$\sigma_{b,p} = \frac{32(1020)}{\pi(0.500)^3} \approx 83.1 \text{ ksi.} \quad (50)$$

The allowable bending stress for the AISI 4140 pin is

$$\sigma_{b,\text{allow}} = \frac{S_y}{n} = \frac{155}{1.5} = 103.3 \text{ ksi.} \quad (51)$$

Since

$$\sigma_{b,p} = 83.1 \text{ ksi} < \sigma_{b,\text{allow}} = 103.3 \text{ ksi}, \quad (52)$$

the selected 0.500 in clevis pin is acceptable in the preliminary bending model when used together with the inward-projecting support sleeves.

The final joint configuration consists of a standard clevis pin, two load-supporting sleeves, a plain washer, and a cotter pin. The sleeves reduce the unsupported pin span while preserving the required bracket spacing. Local contact stresses, sleeve-root stresses, and load distribution through the arm eye should be verified using a refined contact finite-element model after the final joint geometry is completed.

## 7 Scissor Arm Strength and Buckling Calculations

The scissor arms are subjected primarily to axial compression. The maximum compressive force occurs at the minimum operating angle of  $\theta_{\min} = 10.5^\circ$ . The axial force acting on each arm is

$$F_{\text{arm}} = \frac{W}{2 \sin \theta_{\min}} \quad (53)$$

where the rated vehicle load is

$$W = 3307 \text{ lbf.} \quad (54)$$

Therefore,

$$F_{\text{arm}} = \frac{3307}{2 \sin 10.5^\circ} \approx 9070 \text{ lbf.} \quad (55)$$

The arm was modeled as a U-shaped channel consisting of a base plate and two side walls. The principal geometric properties are

$$L = 10.4857 \text{ in,} \quad (56)$$

$$b = 1.8750 \text{ in,} \quad (57)$$

$$h = 0.8000 \text{ in,} \quad (58)$$

$$t = 0.1875 \text{ in,} \quad (59)$$

$$d_h = 0.5000 \text{ in.} \quad (60)$$

The arm material was selected as HSLA Grade 80 steel, with the following material properties:

$$S_y = 80 \text{ ksi,} \quad (61)$$

$$E = 29 \times 10^6 \text{ psi.} \quad (62)$$

A design factor of

$$n = 1.5 \quad (63)$$

was used throughout the calculations.

## 7.1 Cross-Sectional Area and Compressive Stress

The gross cross-sectional area of the U-channel is calculated from the base plate and the two side walls:

$$A = bt + 2t(h - t). \quad (64)$$

Substituting the selected dimensions gives

$$A = (1.875)(0.1875) + 2(0.1875)(0.800 - 0.1875) = 0.5813 \text{ in}^2. \quad (65)$$

The nominal compressive stress in the arm is

$$\sigma_c = \frac{F_{\text{arm}}}{A}. \quad (66)$$

Thus,

$$\sigma_c = \frac{9070}{0.5813} \approx 15.6 \text{ ksi}. \quad (67)$$

The allowable normal stress is

$$\sigma_{\text{allow}} = \frac{S_y}{n} = \frac{80}{1.5} = 53.3 \text{ ksi}. \quad (68)$$

Since

$$\sigma_c = 15.6 \text{ ksi} < \sigma_{\text{allow}} = 53.3 \text{ ksi}, \quad (69)$$

the arm is satisfactory with respect to direct compressive stress.

## 7.2 Global Buckling Check

The minimum second moment of area of the U-channel cross-section was calculated as

$$I_{\min} \approx 0.03044 \text{ in}^4. \quad (70)$$

The corresponding minimum radius of gyration is

$$r_{\min} = \sqrt{\frac{I_{\min}}{A}}. \quad (71)$$

Therefore,

$$r_{\min} = \sqrt{\frac{0.03044}{0.5813}} \approx 0.2288 \text{ in}. \quad (72)$$

The arm was idealized as a pin-ended compression member, and therefore

$$K = 1. \quad (73)$$

The slenderness ratio is

$$\lambda = \frac{KL}{r_{\min}}. \quad (74)$$

Substituting the numerical values gives

$$\lambda = \frac{(1)(10.4857)}{0.2288} \approx 45.8. \quad (75)$$

The limiting slenderness ratio between Johnson and Euler behavior is

$$C_c = \sqrt{\frac{2\pi^2 E}{S_y}}. \quad (76)$$

Hence,

$$C_c = \sqrt{\frac{2\pi^2(29 \times 10^6)}{80 \times 10^3}} \approx 84.6. \quad (77)$$

Since

$$\lambda < C_c, \quad (78)$$

the arm is treated as an intermediate-length column, and the Johnson parabolic relation is used:

$$\sigma_{cr} = S_y \left[ 1 - \frac{S_y}{4\pi^2 E} \left( \frac{KL}{r_{\min}} \right)^2 \right]. \quad (79)$$

The resulting critical buckling stress is

$$\sigma_{cr} \approx 68.3 \text{ ksi}. \quad (80)$$

The corresponding critical buckling load is

$$P_{cr} = \sigma_{cr} A. \quad (81)$$

Thus,

$$P_{cr} = (68.3)(0.5813) \approx 39.7 \text{ kip}. \quad (82)$$

Applying the selected design factor gives the allowable buckling load:

$$P_{\text{allow}} = \frac{P_{cr}}{n} = \frac{39.7}{1.5} \approx 26.5 \text{ kip}. \quad (83)$$

Since

$$F_{\text{arm}} = 9.07 \text{ kip} < P_{\text{allow}} = 26.5 \text{ kip}, \quad (84)$$

the arm is satisfactory with respect to global column buckling.



### 7.3 Net-Section Stress at the Pin Hole

Since the pin passes through both side walls of the U-channel, the net cross-sectional area at the hole location is

$$A_{\text{net}} = A - 2d_h t. \quad (85)$$

Therefore,

$$A_{\text{net}} = 0.5813 - 2(0.500)(0.1875) = 0.3938 \text{ in}^2. \quad (86)$$

The nominal net-section stress is

$$\sigma_{\text{net}} = \frac{F_{\text{arm}}}{A_{\text{net}}}. \quad (87)$$

Thus,

$$\sigma_{\text{net}} = \frac{9070}{0.3938} \approx 23.0 \text{ ksi}. \quad (88)$$

Since

$$\sigma_{\text{net}} = 23.0 \text{ ksi} < \sigma_{\text{allow}} = 53.3 \text{ ksi}, \quad (89)$$

the arm is satisfactory with respect to net-section yielding.

### 7.4 Bearing Stress at the Pin Hole

The arm load is assumed to be distributed equally between the two side walls. Therefore, the load carried by each wall is

$$F_{\text{wall}} = \frac{F_{\text{arm}}}{2} = 4535 \text{ lbf}. \quad (90)$$

The projected bearing stress in each side wall is

$$\sigma_{\text{bearing}} = \frac{F_{\text{wall}}}{d_h t}. \quad (91)$$

Substituting the numerical values gives

$$\sigma_{\text{bearing}} = \frac{4535}{(0.500)(0.1875)} \approx 48.4 \text{ ksi}. \quad (92)$$

Since

$$\sigma_{\text{bearing}} = 48.4 \text{ ksi} < \sigma_{\text{allow}} = 53.3 \text{ ksi}, \quad (93)$$

the bearing region satisfies the selected design factor. However, this location represents one of the more highly stressed regions of the arm.

## 7.5 Shear-Out Check and End-Distance Revision

The initial arm geometry provided a minimum clear distance of

$$e_{\text{initial}} = 0.300 \text{ in} \quad (94)$$

between the edge of the pin hole and the end of the arm.

Each side wall contains two potential shear-out planes. Considering both side walls, the nominal shear-out stress is

$$\tau_{\text{so}} = \frac{F_{\text{arm}}}{4et}. \quad (95)$$

Using the initial clear edge distance gives

$$\tau_{\text{so,initial}} = \frac{9070}{4(0.300)(0.1875)} \approx 40.3 \text{ ksi}. \quad (96)$$

The allowable shear stress, based on the von Mises yield criterion, is

$$\tau_{\text{allow}} = \frac{S_y}{\sqrt{3} n}. \quad (97)$$

Therefore,

$$\tau_{\text{allow}} = \frac{80}{\sqrt{3}(1.5)} \approx 30.8 \text{ ksi}. \quad (98)$$

Since

$$\tau_{\text{so,initial}} > \tau_{\text{allow}}, \quad (99)$$

the initial 0.300 in clear edge distance was found to be insufficient.

The minimum required clear edge distance is obtained from

$$e_{\text{req}} = \frac{F_{\text{arm}}}{4t\tau_{\text{allow}}}. \quad (100)$$

Substituting the numerical values gives

$$e_{\text{req}} = \frac{9070}{4(0.1875)(30.8 \times 10^3)} \approx 0.393 \text{ in}. \quad (101)$$

Accordingly, the arm-end geometry was revised to provide a clear distance of

$$e_{\text{final}} = 0.400 \text{ in}. \quad (102)$$

The revised shear-out stress is

$$\tau_{\text{so,final}} = \frac{9070}{4(0.400)(0.1875)} \approx 30.2 \text{ ksi}. \quad (103)$$

Since

$$\tau_{\text{so,final}} = 30.2 \text{ ksi} < \tau_{\text{allow}} = 30.8 \text{ ksi}, \quad (104)$$

the revised geometry satisfies the selected design requirement.

The corresponding minimum distance from the hole center to the arm end is

$$e_c = e_{\text{final}} + \frac{d_h}{2}. \quad (105)$$

Thus,

$$e_c = 0.400 + \frac{0.500}{2} = 0.650 \text{ in.} \quad (106)$$

The final arm geometry therefore uses a minimum hole-center-to-end distance of 0.650 in.

## 8 Power Screw Design

The lifting mechanism is actuated by a single-start ACME power screw. One side of the mechanism contains a threaded trunnion nut, while the opposite side supports the rotating screw and transfers the axial reaction through an axial support arrangement.

A right-hand 3/4-6 ACME thread was selected for the power screw. The principal thread dimensions used in the preliminary calculations are

$$d = 0.750 \text{ in,} \quad (107)$$

$$p = \frac{1}{6} = 0.1667 \text{ in,} \quad (108)$$

$$l = 0.1667 \text{ in,} \quad (109)$$

$$d_m \approx 0.667 \text{ in,} \quad (110)$$

$$d_r \approx 0.583 \text{ in,} \quad (111)$$

where  $d$  is the nominal major diameter,  $p$  is the thread pitch,  $l$  is the lead,  $d_m$  is the approximate mean thread diameter, and  $d_r$  is the approximate root diameter. Since a single-start thread is used, the lead is equal to the pitch.

The screw was assumed to be manufactured from heat-treated AISI 4140 steel with an approximate yield strength of

$$S_y = 155 \text{ ksi.} \quad (112)$$

A design factor of

$$n = 1.5 \quad (113)$$

was used for the preliminary strength calculations.

### 8.1 Maximum Axial Screw Load

The maximum axial force in the power screw occurs at the minimum operating angle of the scissor mechanism. The horizontal components of the upper and lower arm forces act on the threaded trunnion block and produce the axial screw load.

The axial force in one scissor arm is

$$F_{\text{arm}} = \frac{W}{2 \sin \theta}. \quad (114)$$

The corresponding axial force acting through the screw is

$$F_s = 2F_{\text{arm}} \cos \theta. \quad (115)$$

Substitution of the arm-force expression gives

$$F_s = W \cot \theta. \quad (116)$$

Using the rated load

$$W = 3307 \text{ lbf} \quad (117)$$

and the minimum operating angle

$$\theta_{\min} = 10.5^\circ, \quad (118)$$

the maximum axial screw load is

$$F_{s,\max} = 3307 \cot 10.5^\circ \approx 17.8 \times 10^3 \text{ lbf.} \quad (119)$$

Therefore,

$$F_{s,\max} \approx 17.8 \text{ kip.} \quad (120)$$

This load is used for the screw stress and operating-torque calculations.

## 8.2 Axial Stress at the Screw Root

The minimum load-carrying area of the screw occurs at the thread root. The root area is approximated as

$$A_r = \frac{\pi d_r^2}{4}. \quad (121)$$

Substituting the approximate root diameter gives

$$A_r = \frac{\pi(0.583)^2}{4} \approx 0.267 \text{ in}^2. \quad (122)$$

The corresponding nominal axial tensile stress is

$$\sigma_s = \frac{F_{s,\max}}{A_r}. \quad (123)$$

Therefore,

$$\sigma_s = \frac{17.8 \times 10^3}{0.267} \approx 66.7 \text{ ksi.} \quad (124)$$

The allowable normal stress is

$$\sigma_{\text{allow}} = \frac{S_y}{n} = \frac{155}{1.5} = 103.3 \text{ ksi.} \quad (125)$$

Since

$$\sigma_s = 66.7 \text{ ksi} < \sigma_{\text{allow}} = 103.3 \text{ ksi}, \quad (126)$$

the selected screw diameter is satisfactory with respect to direct axial stress.

### 8.3 Lead Angle

The lead angle of the power screw is calculated at the mean thread diameter:

$$\tan \lambda = \frac{l}{\pi d_m}. \quad (127)$$

Substituting the selected thread dimensions gives

$$\tan \lambda = \frac{0.1667}{\pi(0.667)} \approx 0.0795. \quad (128)$$

Therefore,

$$\lambda = \tan^{-1}(0.0795) \approx 4.55^\circ. \quad (129)$$

The relatively small lead angle provides a high mechanical advantage and supports the self-locking behavior required for a manually operated vehicle jack.

### 8.4 Self-Locking Check

For an ACME thread, the thread-angle effect is included by defining an equivalent coefficient of friction:

$$\mu' = \frac{\mu}{\cos \alpha}, \quad (130)$$

where  $\alpha$  is one-half of the included ACME thread angle. For a standard ACME thread,

$$\alpha = 14.5^\circ. \quad (131)$$

Assuming a preliminary steel–bronze thread friction coefficient of

$$\mu = 0.15, \quad (132)$$

the equivalent coefficient of friction becomes

$$\mu' = \frac{0.15}{\cos 14.5^\circ} \approx 0.155. \quad (133)$$

The corresponding equivalent friction angle is

$$\phi' = \tan^{-1}(\mu'). \quad (134)$$

Thus,

$$\phi' = \tan^{-1}(0.155) \approx 8.8^\circ. \quad (135)$$

A power screw is considered self-locking when

$$\lambda < \phi'. \quad (136)$$

For the selected screw,

$$4.55^\circ < 8.8^\circ. \quad (137)$$

Therefore, the selected 3/4-6 ACME power screw is self-locking under the assumed friction conditions. This prevents the screw from being back-driven by the applied vehicle load during normal operation.

The exact root and mean thread diameters should be updated using the final NX thread-table geometry before the design is released for manufacture. The operating torque, torsional stress, combined von Mises stress, and axial-support design are evaluated in the following subsections.



## 8.5 Raising Torque

The torque required to raise the load through the ACME thread is calculated using the equivalent friction angle approach:

$$T_{\text{raise}} = F_{s,\text{max}} \frac{d_m}{2} \tan(\lambda + \phi'). \quad (138)$$

Using

$$F_{s,\text{max}} = 17.8 \times 10^3 \text{ lbf}, \quad (139)$$

$$d_m = 0.667 \text{ in}, \quad (140)$$

$$\lambda = 4.55^\circ, \quad (141)$$

$$\phi' = 8.81^\circ, \quad (142)$$

the required thread torque becomes

$$T_{\text{raise}} = (17.8 \times 10^3) \frac{0.667}{2} \tan(4.55^\circ + 8.81^\circ). \quad (143)$$

Therefore,

$$T_{\text{raise}} \approx 1409 \text{ lbf} \cdot \text{in}. \quad (144)$$

Equivalently,

$$T_{\text{raise}} \approx 117.4 \text{ lbf} \cdot \text{ft}. \quad (145)$$

This value represents the torque required to overcome friction in the ACME thread only. Additional torque associated with the axial support bearing or thrust surface must be added after the final axial-support arrangement is selected.

## 8.6 Lowering Torque

Because the selected thread is self-locking, a finite reverse torque is required to lower the load. The lowering torque is

$$T_{\text{lower}} = F_{s,\text{max}} \frac{d_m}{2} \tan(\phi' - \lambda). \quad (146)$$

Substituting the selected values gives

$$T_{\text{lower}} = (17.8 \times 10^3) \frac{0.667}{2} \tan(8.81^\circ - 4.55^\circ). \quad (147)$$

Thus,

$$T_{\text{lower}} \approx 442 \text{ lbf} \cdot \text{in.} \quad (148)$$

Equivalently,

$$T_{\text{lower}} \approx 36.8 \text{ lbf} \cdot \text{ft.} \quad (149)$$

Since the lowering torque is positive, the load cannot back-drive the screw under the assumed friction conditions.

## 8.7 Torsional Stress at the Screw Root

The maximum torsional stress occurs at the minimum screw diameter. For a solid circular section, the torsional shear stress is

$$\tau_t = \frac{16T_{\text{raise}}}{\pi d_r^3}. \quad (150)$$

Using the root diameter

$$d_r = 0.583 \text{ in,} \quad (151)$$

the torsional shear stress becomes

$$\tau_t = \frac{16(1409)}{\pi(0.583)^3}. \quad (152)$$

Therefore,

$$\tau_t \approx 36.2 \text{ ksi.} \quad (153)$$

## 8.8 Combined Axial and Torsional Stress

The screw is subjected simultaneously to axial tension and torsion. The previously calculated axial stress is

$$\sigma_s = 66.7 \text{ ksi.} \quad (154)$$

The equivalent von Mises stress is calculated as

$$\sigma_{\text{vm}} = \sqrt{\sigma_s^2 + 3\tau_t^2}. \quad (155)$$

Substituting the calculated stresses gives

$$\sigma_{\text{vm}} = \sqrt{(66.7)^2 + 3(36.2)^2}. \quad (156)$$

Thus,

$$\sigma_{\text{vm}} \approx 91.6 \text{ ksi.} \quad (157)$$

The allowable normal stress for the heat-treated AISI 4140 screw is

$$\sigma_{\text{allow}} = \frac{S_y}{n} = \frac{155}{1.5} = 103.3 \text{ ksi.} \quad (158)$$

Since

$$\sigma_{\text{vm}} = 91.6 \text{ ksi} < \sigma_{\text{allow}} = 103.3 \text{ ksi}, \quad (159)$$

the selected screw satisfies the preliminary combined axial and torsional stress requirement.

The resulting design factor based on the assumed screw yield strength is

$$n_{\text{actual}} = \frac{S_y}{\sigma_{\text{vm}}} = \frac{155}{91.6} \approx 1.69. \quad (160)$$

## 8.9 Axial Loading Condition and Buckling Consideration

Under the applied vehicle load, the horizontal components of the scissor-arm forces tend to push the two trunnion blocks away from each other. The power screw restrains this separating motion and therefore acts as a tension member between the fixed support block and the moving threaded nut block.

Accordingly, the previously calculated axial stress in the screw represents a tensile stress:

$$\sigma_s = \frac{F_{s,\text{max}}}{A_r} = 66.7 \text{ ksi}. \quad (161)$$

Since the load-carrying portion of the screw is subjected primarily to axial tension rather than axial compression, global column buckling does not govern the screw design during normal lifting operation.

The maximum distance between the two side-pivot centers occurs at the minimum operating angle and is

$$x_{\text{max}} = 2L \cos \theta_{\text{min}}. \quad (162)$$

Using

$$L = 10.4857 \text{ in}, \quad (163)$$

$$\theta_{\text{min}} = 10.5^\circ, \quad (164)$$

the maximum screw span between the trunnion-block centers is

$$x_{\text{max}} = 2(10.4857) \cos 10.5^\circ \approx 20.62 \text{ in}. \quad (165)$$

Although this span would be relatively slender if subjected to compression, the service load produces tension in the screw and therefore stabilizes it against global buckling.

Local bending may still arise from manufacturing tolerances, misalignment, clearance at the trunnion blocks, or eccentric force transmission. These secondary effects are not included in the preliminary analytical model and should be assessed through assembly alignment control and, if required, a refined finite-element analysis of the completed screw and support geometry.

## 9 Threaded Trunnion Nut Design

The moving trunnion block contains the internal ACME thread and converts the rotation of the power screw into linear motion. The nut was designed using a bronze alloy in order to reduce friction, avoid adhesive wear between similar materials, and provide a replaceable sacrificial component.

For the preliminary calculations, C95400 aluminum bronze was assumed with an approximate yield strength of

$$S_{y,n} = 35 \text{ ksi.} \quad (166)$$

The nut contains a right-hand 3/4-6 ACME internal thread. The selected thread engagement length is

$$L_e = 1.75 \text{ in.} \quad (167)$$

The thread pitch is

$$p = \frac{1}{6} = 0.1667 \text{ in.} \quad (168)$$

Therefore, the approximate number of engaged threads is

$$N_e = \frac{L_e}{p}. \quad (169)$$

Thus,

$$N_e = \frac{1.75}{0.1667} \approx 10.5. \quad (170)$$

Approximately ten full threads are therefore engaged during operation.

## 9.1 Average Thread Bearing Pressure

The axial load is transferred between the screw and nut through the bearing surfaces of the ACME threads. Assuming that one-half of the axial pitch contributes to the projected bearing area, the total projected bearing area is approximated by

$$A_{b,t} = \frac{\pi d_m L_e}{2}, \quad (171)$$

where  $d_m$  is the mean thread diameter.

Using

$$d_m = 0.667 \text{ in}, \quad (172)$$

the projected bearing area becomes

$$A_{b,t} = \frac{\pi(0.667)(1.75)}{2} \approx 1.833 \text{ in}^2. \quad (173)$$

The average thread bearing pressure is therefore

$$p_b = \frac{F_{s,\max}}{A_{b,t}}. \quad (174)$$

Substituting the maximum screw load gives

$$p_b = \frac{17.8 \times 10^3}{1.833} \approx 9.71 \text{ ksi}. \quad (175)$$

The corresponding bearing design factor based on the assumed bronze yield strength is

$$n_b = \frac{S_{y,n}}{p_b}. \quad (176)$$

Thus,

$$n_b = \frac{35}{9.71} \approx 3.60. \quad (177)$$

The average thread bearing pressure is therefore acceptable for the selected preliminary bronze alloy.

## 9.2 Nut-Thread Shear Check

A possible failure mode of the threaded nut is shearing of the internal thread material. The approximate nut-thread shear area is

$$A_{\text{sh},n} = \frac{\pi d L_e}{2}, \quad (178)$$

where  $d$  is the nominal major thread diameter.

Therefore,

$$A_{\text{sh},n} = \frac{\pi(0.750)(1.75)}{2} \approx 2.062 \text{ in}^2. \quad (179)$$

The average shear stress in the nut threads is

$$\tau_n = \frac{F_{s,\text{max}}}{A_{\text{sh},n}}. \quad (180)$$

Thus,

$$\tau_n = \frac{17.8 \times 10^3}{2.062} \approx 8.63 \text{ ksi}. \quad (181)$$

Using the von Mises shear-yield approximation, the allowable shear stress for the bronze nut is

$$\tau_{\text{allow},n} = \frac{S_{y,n}}{\sqrt{3}n}. \quad (182)$$

For a design factor of  $n = 1.5$ ,

$$\tau_{\text{allow},n} = \frac{35}{\sqrt{3}(1.5)} \approx 13.5 \text{ ksi.} \quad (183)$$

Since

$$\tau_n = 8.63 \text{ ksi} < \tau_{\text{allow},n} = 13.5 \text{ ksi}, \quad (184)$$

the nut threads are satisfactory with respect to average thread shear.

### 9.3 Screw-Thread Shear Check

The external screw threads were also checked against stripping. The approximate screw-thread shear area is

$$A_{\text{sh},s} = \frac{\pi d_r L_e}{2}, \quad (185)$$

where  $d_r$  is the screw root diameter.

Using

$$d_r = 0.583 \text{ in}, \quad (186)$$

the screw-thread shear area becomes

$$A_{\text{sh},s} = \frac{\pi(0.583)(1.75)}{2} \approx 1.602 \text{ in}^2. \quad (187)$$

The resulting average screw-thread shear stress is

$$\tau_s = \frac{F_{s,\text{max}}}{A_{\text{sh},s}}. \quad (188)$$

Therefore,

$$\tau_s = \frac{17.8 \times 10^3}{1.602} \approx 11.1 \text{ ksi.} \quad (189)$$



The allowable shear stress for the heat-treated AISI 4140 screw is

$$\tau_{\text{allow},s} = \frac{S_y}{\sqrt{3}n}. \quad (190)$$

Thus,

$$\tau_{\text{allow},s} = \frac{155}{\sqrt{3}(1.5)} \approx 59.7 \text{ ksi}. \quad (191)$$

Since

$$\tau_s = 11.1 \text{ ksi} < \tau_{\text{allow},s} = 59.7 \text{ ksi}, \quad (192)$$

thread stripping of the steel screw does not govern the design.

#### 9.4 Preliminary Nut Dimensions

The threaded section was incorporated into a rectangular central nut body. The original integral bronze trunnion concept was evaluated and subsequently replaced by a separate steel sleeve and through-pin assembly. The preliminary dimensions are

$$L_n = 1.75 \text{ in}, \quad (193)$$

$$b_n = 1.00 \text{ in}, \quad (194)$$

$$h_n = 1.00 \text{ in}, \quad (195)$$

$$d_t = 0.875 \text{ in}, \quad (196)$$

where  $L_n$  is the threaded body length,  $b_n$  is the body width,  $h_n$  is the body height, and  $d_t$  is the trunnion diameter.

A transition fillet in the range of

$$r_f = 0.10\text{--}0.125 \text{ in} \quad (197)$$

is provided between each trunnion and the central nut body in order to reduce the local stress concentration.

The analytical calculations assume uniform load distribution among the engaged threads. In practice, the first few threads carry a larger portion of the load because of elastic deformation and manufacturing tolerances. Consequently, the final nut geometry, trunnion-root region, and local thread stresses should be reviewed using the completed CAD geometry and, if necessary, a local finite-element contact analysis.

## 9.5 Trunnion Bearing and Bending Check

The threaded nut block transfers the forces from the scissor arms through two cylindrical trunnions located on opposite sides of the central nut body. The preliminary trunnion diameter and projection length were selected as

$$d_t = 0.875 \text{ in}, \quad (198)$$

$$L_t = 0.325 \text{ in}. \quad (199)$$

The maximum horizontal force acting on the threaded nut block is equal to the maximum axial screw load:

$$F_{s,\max} = 17.8 \text{ kip}. \quad (200)$$

Assuming that this load is distributed equally between the two trunnions, the radial load carried by one trunnion is

$$F_t = \frac{F_{s,\max}}{2}. \quad (201)$$

Therefore,

$$F_t = \frac{17.8}{2} = 8.90 \text{ kip}. \quad (202)$$

### 9.5.1 Bearing Stress at the Trunnion–Arm Interface

The projected bearing area between one trunnion and one arm wall is

$$A_{b,t} = d_t t, \quad (203)$$

where the arm-wall thickness is

$$t = 0.1875 \text{ in.} \quad (204)$$

The nominal bearing stress is therefore

$$\sigma_{b,t} = \frac{F_t}{d_t t}. \quad (205)$$

Substituting the preliminary dimensions gives

$$\sigma_{b,t} = \frac{8.90 \times 10^3}{(0.875)(0.1875)} \approx 54.2 \text{ ksi.} \quad (206)$$

For the preliminary C95400 aluminum-bronze nut material, the allowable normal stress based on a yield strength of 35 ksi and a design factor of 1.5 is

$$\sigma_{\text{allow},n} = \frac{35}{1.5} = 23.3 \text{ ksi.} \quad (207)$$

Since

$$\sigma_{b,t} = 54.2 \text{ ksi} > \sigma_{\text{allow},n} = 23.3 \text{ ksi}, \quad (208)$$

the preliminary integral bronze trunnion does not satisfy the selected bearing-stress requirement under the conservative load-distribution assumption.

### 9.5.2 Nominal Trunnion Bending Stress

The trunnion was conservatively idealized as a short cantilever extending from the central nut body. The maximum bending moment occurs at the trunnion root:

$$M_t = F_t a, \quad (209)$$

where  $a$  is the distance between the nut-body surface and the resultant bearing-load location.

For the most conservative preliminary estimate, the load was assumed to act at the outer end of the trunnion:

$$a = L_t = 0.325 \text{ in.} \quad (210)$$

Thus,

$$M_t = (8.90 \times 10^3)(0.325) \approx 2893 \text{ lbf} \cdot \text{in.} \quad (211)$$

The nominal bending stress at the root of a solid circular trunnion is

$$\sigma_{b,\text{nom}} = \frac{32M_t}{\pi d_t^3}. \quad (212)$$

Therefore,

$$\sigma_{b,\text{nom}} = \frac{32(2893)}{\pi(0.875)^3} \approx 44.0 \text{ ksi.} \quad (213)$$

Even before accounting for local geometric stress concentration, the nominal bending stress exceeds the allowable stress of the preliminary bronze material.

### 9.5.3 Stress Concentration at the Trunnion Root

The transition between the cylindrical trunnion and the rectangular nut body introduces a local geometric discontinuity. Consequently, the maximum local bending stress is greater than the nominal beam-theory result.

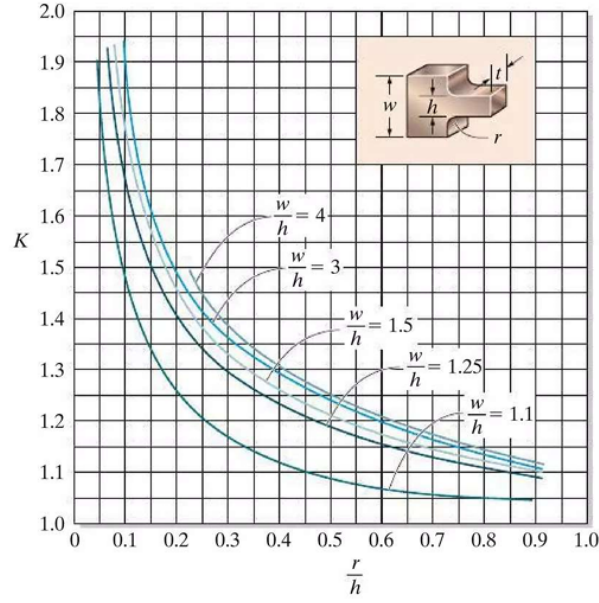


Figure 3: Stress Concentration Factor

The local peak stress may be expressed as

$$\sigma_{b,\max} = K_t \sigma_{b,\text{nom}}, \quad (214)$$

where  $K_t$  is the theoretical stress-concentration factor associated with the trunnion diameter, body dimensions, shoulder geometry, and root fillet radius.

A root fillet in the range

$$r_f = 0.10\text{--}0.125 \text{ in} \quad (215)$$

was included in the preliminary geometry. However, the exact value of  $K_t$  cannot be determined accurately from the nominal dimensions alone, because the trunnion blends into a non-axisymmetric rectangular body.

For illustration, if a preliminary theoretical stress-concentration factor of

$$K_t = 1.5 \quad (216)$$

is assumed, the estimated local bending stress becomes

$$\sigma_{b,\max} = 1.5(44.0) = 66.0 \text{ ksi.} \quad (217)$$

This value is substantially greater than the yield strength of the preliminary bronze alloy. The result confirms that the trunnion-root transition, rather than the threaded portion of the nut, governs the preliminary integral-trunnion design.

The value  $K_t = 1.5$  is used only as a preliminary illustration and is not treated as a final design value. The actual stress concentration should be obtained from an appropriate shoulder-filleting correlation or from a refined finite-element analysis of the completed three-dimensional geometry.

#### 9.5.4 Trunnion Shear Stress

The average direct shear stress in one trunnion is

$$\tau_{t,\text{avg}} = \frac{F_t}{A_t}, \quad (218)$$

where

$$A_t = \frac{\pi d_t^2}{4}. \quad (219)$$

Therefore,

$$A_t = \frac{\pi(0.875)^2}{4} \approx 0.601 \text{ in}^2, \quad (220)$$

and

$$\tau_{t,\text{avg}} = \frac{8.90 \times 10^3}{0.601} \approx 14.8 \text{ ksi.} \quad (221)$$

For a circular cross-section, the maximum transverse shear stress is approximately

$$\tau_{t,\max} = \frac{4}{3}\tau_{t,\text{avg}}. \quad (222)$$

Thus,

$$\tau_{t,\max} \approx \frac{4}{3}(14.8) \approx 19.7 \text{ ksi.} \quad (223)$$

The direct shear stress alone is lower than the nominal bending stress; however, it contributes to the combined stress state at the trunnion root.

A preliminary von Mises estimate may be written as

$$\sigma_{\text{vm},t} = \sqrt{(K_t \sigma_{b,\text{nom}})^2 + 3\tau_{t,\max}^2}. \quad (224)$$

Using  $K_t = 1.5$  gives

$$\sigma_{\text{vm},t} = \sqrt{(66.0)^2 + 3(19.7)^2} \approx 74.3 \text{ ksi.} \quad (225)$$

This result further demonstrates that the preliminary integral bronze trunnion is not acceptable under the conservative loading model.

### 9.5.5 Design Revision

Based on the bearing, bending, and local stress-concentration results, the integral bronze trunnion configuration was not retained as the final load-carrying solution.

The revised design separates the functions of the threaded nut and the pivot pin:

- the bronze nut body retains the internal ACME thread and provides the sliding thread interface;
- a separate high-strength steel trunnion pin transfers the arm forces;
- the steel pin passes through the bronze nut body and is supported on both sides;
- generous fillets or fitted steel sleeves are used at the load-transfer regions;
- the bronze component is no longer required to carry the full cantilever bending load through an integral cylindrical projection.

This revision substantially reduces the bending and stress-concentration demands on the bronze nut body while allowing the pivot component to be manufactured from a higher-strength and more wear-resistant steel.

The final steel trunnion-pin diameter, bearing length, and fit should be verified after the revised nut-block geometry is completed. Local contact pressure and stress concentration around the transverse pin bore should also be evaluated using the final CAD model.

## 9.6 Revised Steel Trunnion Assembly

The initial integral bronze trunnion was replaced by a separate steel trunnion assembly. In the revised configuration, a steel bearing sleeve passes through the bronze threaded nut body, while a high-strength steel through-pin retains the complete assembly.

The principal dimensions of the revised trunnion assembly are

$$d_{\text{pin}} = 0.625 \text{ in}, \quad (226)$$

$$d_{\text{sleeve}} = 0.875 \text{ in}, \quad (227)$$

$$t_{\text{eff}} = 0.200 \text{ in}, \quad (228)$$

$$b_n = 1.00 \text{ in}, \quad (229)$$

where  $d_{\text{pin}}$  is the through-pin diameter,  $d_{\text{sleeve}}$  is the external bearing diameter of the steel sleeve,  $t_{\text{eff}}$  is the locally reinforced effective arm-wall thickness, and  $b_n$  is the width of the bronze nut body.

The through-pin was assumed to be manufactured from heat-treated AISI 4140 steel. The external sleeve provides the bearing surface for the arm and also distributes the load through the bronze nut body.

The nominal sleeve wall thickness is

$$t_{\text{sleeve}} = \frac{d_{\text{sleeve}} - d_{\text{pin}}}{2}. \quad (230)$$

Therefore,

$$t_{\text{sleeve}} = \frac{0.875 - 0.625}{2} = 0.125 \text{ in}. \quad (231)$$



### 9.6.1 Arm-to-Sleeve Bearing Stress

The total horizontal load acting on the threaded nut block is

$$F_{s,\max} = 17.8 \text{ kip.} \quad (232)$$

Assuming equal load distribution between the two sides of the trunnion assembly, the load carried at each arm-to-sleeve interface is

$$F_t = \frac{F_{s,\max}}{2}. \quad (233)$$

Thus,

$$F_t = \frac{17.8}{2} = 8.90 \text{ kip.} \quad (234)$$

The projected bearing stress at one arm-to-sleeve interface is

$$\sigma_{b,\text{arm}} = \frac{F_t}{d_{\text{sleeve}} t_{\text{eff}}}. \quad (235)$$

Substituting the revised dimensions gives

$$\sigma_{b,\text{arm}} = \frac{8.90 \times 10^3}{(0.875)(0.200)} \approx 50.9 \text{ ksi.} \quad (236)$$

For the HSLA Grade 80 arm material, the allowable normal stress is

$$\sigma_{\text{allow},\text{arm}} = \frac{S_{y,\text{arm}}}{n}. \quad (237)$$

Using

$$S_{y,\text{arm}} = 80 \text{ ksi,} \quad (238)$$

$$n = 1.5, \quad (239)$$

the allowable stress becomes

$$\sigma_{\text{allow,arm}} = \frac{80}{1.5} = 53.3 \text{ ksi.} \quad (240)$$

Since

$$\sigma_{\text{b,arm}} = 50.9 \text{ ksi} < \sigma_{\text{allow,arm}} = 53.3 \text{ ksi}, \quad (241)$$

the locally reinforced arm region satisfies the selected bearing-stress requirement.

The result also shows that the arm-to-sleeve bearing region remains highly loaded. Consequently, the local reinforcement must extend sufficiently around the hole to avoid an abrupt stiffness transition.

### 9.6.2 Steel Through-Pin Shear Check

The through-pin transfers the trunnion load through two effective shear planes. The cross-sectional area of one shear plane is

$$A_{\text{pin}} = \frac{\pi d_{\text{pin}}^2}{4}. \quad (242)$$

Therefore,

$$A_{\text{pin}} = \frac{\pi(0.625)^2}{4} \approx 0.3068 \text{ in}^2. \quad (243)$$

The nominal double-shear stress in the pin is

$$\tau_{\text{pin}} = \frac{F_{s,\text{max}}}{2A_{\text{pin}}}. \quad (244)$$

Substituting the maximum load gives

$$\tau_{\text{pin}} = \frac{17.8 \times 10^3}{2(0.3068)} \approx 29.0 \text{ ksi.} \quad (245)$$

For heat-treated AISI 4140 steel with

$$S_{y,\text{pin}} = 155 \text{ ksi}, \quad (246)$$

the allowable shear stress based on the von Mises yield criterion is

$$\tau_{\text{allow},\text{pin}} = \frac{S_{y,\text{pin}}}{\sqrt{3}n}. \quad (247)$$

Therefore,

$$\tau_{\text{allow},\text{pin}} = \frac{155}{\sqrt{3}(1.5)} \approx 59.7 \text{ ksi}. \quad (248)$$

Since

$$\tau_{\text{pin}} = 29.0 \text{ ksi} < \tau_{\text{allow},\text{pin}} = 59.7 \text{ ksi}, \quad (249)$$

the selected 0.625 in through-pin is satisfactory with respect to direct double shear.

### 9.6.3 Bearing Stress in the Bronze Nut Body

The steel sleeve passes through the complete width of the bronze nut body. The projected bearing area between the sleeve and the nut body is approximated as

$$A_{\text{b,nut}} = d_{\text{sleeve}}b_n. \quad (250)$$

Therefore,

$$A_{\text{b,nut}} = (0.875)(1.00) = 0.875 \text{ in}^2. \quad (251)$$

The corresponding nominal bearing stress in the bronze body is

$$\sigma_{\text{b,nut}} = \frac{F_{s,\text{max}}}{A_{\text{b,nut}}}. \quad (252)$$

Thus,

$$\sigma_{b,nut} = \frac{17.8 \times 10^3}{0.875} \approx 20.3 \text{ ksi.} \quad (253)$$

For the preliminary C95400 aluminum-bronze material, the allowable normal stress is

$$\sigma_{allow,nut} = \frac{S_{y,n}}{n} = \frac{35}{1.5} = 23.3 \text{ ksi.} \quad (254)$$

Since

$$\sigma_{b,nut} = 20.3 \text{ ksi} < \sigma_{allow,nut} = 23.3 \text{ ksi}, \quad (255)$$

the revised steel sleeve provides sufficient nominal bearing area within the bronze nut body.

#### 9.6.4 Local Bending and Stress-Concentration Considerations

Although the direct shear and nominal bearing checks are satisfactory, the trunnion assembly contains several geometric discontinuities that may produce local stress concentrations.

The most critical regions include

- the transition between the reinforced arm-eye region and the original arm wall;
- the ends of the steel sleeve where the arm load enters the trunnion assembly;
- the edges of the transverse sleeve bore in the bronze nut body;
- any shoulder, retaining groove, or diameter transition machined into the steel through-pin;
- local clearance between the through-pin, sleeve, and bronze nut body.

The local peak stress near a geometric discontinuity may be represented by

$$\sigma_{max} = K_t \sigma_{nom}, \quad (256)$$

where  $K_t$  is the theoretical stress-concentration factor associated with the actual local geometry.

The present analytical calculations use nominal projected bearing and beam-theory stresses. They do not include the complete three-dimensional contact-pressure distribution or the local increase in stress caused by bore edges, sleeve ends, retaining grooves, or abrupt reinforcement transitions.

Accordingly, the final design should incorporate

- smooth transitions between the locally reinforced and nominal arm thicknesses;
- generous fillets at all sleeve-support and reinforcement transitions;
- avoidance of retaining grooves within highly loaded shear or bearing regions;
- close control of pin-to-sleeve and sleeve-to-body clearances;
- hardened and adequately lubricated steel bearing surfaces.

Because the exact stress-concentration factor depends on the completed three-dimensional geometry, a local nonlinear contact finite-element analysis is recommended for the final trunnion assembly. Particular attention should be given to sleeve-end bending, bronze-body bore deformation, and peak stresses at the local arm reinforcement.

The revised configuration is retained because it separates the low-friction threaded function of the bronze nut from the high-load bearing and shear functions of the steel trunnion components.

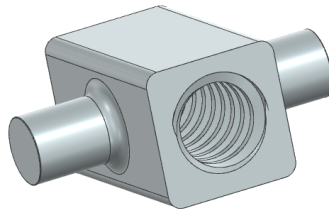


Figure 4: Nut Block

## 10 Axial Screw Support Design

In the adopted screw arrangement, one trunnion block contains the moving threaded nut, while the opposite trunnion block supports the rotating power screw. The support-side block allows screw rotation while preventing axial translation of the screw.

The axial reaction transmitted through this support is approximately equal to the maximum power-screw load:

$$F_a = F_{s,\max} = 17.8 \text{ kip.} \quad (257)$$

In SI units,

$$F_a = 17.8 \times 4.448 \approx 79.2 \text{ kN.} \quad (258)$$

Because this load is considerably greater than the capacity of compact thrust ball bearings having a bore close to the nominal screw diameter, an axial cylindrical roller bearing was selected.

## 10.1 Selected Axial Roller Bearing

The preliminary axial support bearing was selected as

$$\text{SKF 81206 TN.} \quad (259)$$

This is a single-direction cylindrical roller thrust bearing. Its principal dimensions and catalogue ratings are

$$d_b = 30 \text{ mm,} \quad (260)$$

$$D_b = 52 \text{ mm,} \quad (261)$$

$$T_b = 16 \text{ mm,} \quad (262)$$

$$C_a \approx 50 \text{ kN,} \quad (263)$$

$$C_{0a} \approx 134 \text{ kN,} \quad (264)$$

where  $d_b$  is the bearing bore diameter,  $D_b$  is the outside diameter,  $T_b$  is the total bearing thickness,  $C_a$  is the basic dynamic axial load rating, and  $C_{0a}$  is the basic static axial load rating.

## 10.2 Static Load Check

The equivalent static axial bearing load is taken as

$$P_{0a} = F_a = 79.2 \text{ kN.} \quad (265)$$

The preliminary static safety factor is

$$s_0 = \frac{C_{0a}}{P_{0a}}. \quad (266)$$

Therefore,

$$s_0 = \frac{134}{79.2} \approx 1.69. \quad (267)$$

Since

$$s_0 > 1.5, \quad (268)$$

the selected bearing satisfies the preliminary static-capacity requirement.

The dynamic load rating is lower than the maximum applied axial load. However, the jack is operated at extremely low rotational speed and for short, intermittent lifting cycles. Therefore, the static load rating is considered more relevant than conventional high-cycle bearing-life calculations for the preliminary selection.

The bearing should nevertheless be inspected for permanent deformation, roller indentation, and washer damage if the jack is subjected to overload or impact loading.

### 10.3 Bearing-to-Screw Adapter

The nominal power-screw diameter is

$$d_s = 0.750 \text{ in} = 19.05 \text{ mm}, \quad (269)$$

whereas the selected bearing has a bore diameter of

$$d_b = 30 \text{ mm.} \quad (270)$$

Consequently, the bearing cannot be located directly on the threaded screw section. A steel adapter sleeve and thrust collar are introduced between the screw and the bearing.

The adapter sleeve has the preliminary diameters

$$d_{\text{adapter,in}} \approx 19.05 \text{ mm}, \quad (271)$$

$$d_{\text{adapter,out}} = 30.00 \text{ mm}. \quad (272)$$

The nominal radial wall thickness of the adapter is therefore

$$t_{\text{adapter}} = \frac{d_{\text{adapter,out}} - d_{\text{adapter,in}}}{2}. \quad (273)$$

Thus,

$$t_{\text{adapter}} = \frac{30.00 - 19.05}{2} \approx 5.48 \text{ mm}. \quad (274)$$

The adapter sleeve centers the bearing washer on the power screw. However, the sleeve must not be relied upon solely through friction to transmit the full axial load. The axial reaction must instead pass through a positive mechanical shoulder or thrust collar connected to the screw.

The intended load path is

$$\text{power screw} \rightarrow \text{thrust collar} \rightarrow \text{bearing shaft washer} \rightarrow \text{rollers} \rightarrow \text{housing washer} \rightarrow \text{support trunnion block}. \quad (275)$$

## 10.4 Support-Block Arrangement

The bearing is located inside the non-threaded support-side trunnion block. This block moves with the scissor linkage but remains fixed relative to the axial position of the screw.

The support block performs the following functions:

- supports and centers the rotating screw;
- prevents axial translation of the screw;



- transfers the axial reaction from the screw into the scissor arms;
- provides a seat for the stationary bearing washer;
- protects the roller assembly from contamination and misalignment.

The housing seat must provide full circumferential support to the stationary bearing washer. The rotating washer must bear against a continuous hardened thrust collar rather than against the threaded portion of the screw.

The bearing arrangement should include a retaining element on the opposite side of the support block so that the screw remains axially located when the jack is unloaded or handled. This retaining element may consist of a locknut, retaining collar, or mechanically secured end cap.

## 10.5 Stress-Concentration and Contact Considerations

The transition from the 0.750 in threaded screw section to the larger thrust-collar or adapter region introduces a geometric discontinuity. The nominal axial and torsional stresses calculated for the thread root may therefore be locally amplified.

The peak stress at the shoulder may be expressed as

$$\sigma_{\max} = K_t \sigma_{\text{nom}}, \quad (276)$$

where  $K_t$  depends on the diameter ratio, shoulder radius, groove geometry, and local loading condition.

To reduce the stress concentration, the screw-support geometry should include

- a generous shoulder fillet;
- no retaining groove within the highly loaded shoulder region;
- a hardened and ground thrust-collar face;
- full washer support without edge loading;
- accurate axial alignment between the screw, bearing, and support block.

The bearing washers must remain parallel under load. Misalignment may cause nonuniform roller loading, local indentation, and substantially increased contact stress.

The final shoulder dimensions, adapter connection, housing-seat thickness, and retaining arrangement should be verified using the completed CAD geometry. A local contact finite-element analysis is recommended for the screw shoulder, bearing-seat region, and support-block load path.

## 11 Manual Handle Design

The power screw is operated manually through a removable crank handle. The handle must provide sufficient mechanical advantage to generate the required raising torque while remaining strong enough to resist bending during operation.

The previously calculated thread-raising torque is

$$T_{\text{raise}} = 1409 \text{ lbf} \cdot \text{in.} \quad (277)$$

This value includes friction within the ACME thread but does not include additional friction associated with the axial support bearing, seals, misalignment, or manufacturing tolerances. Therefore, the resulting handle force represents a preliminary lower-bound estimate.

### 11.1 Required Handle Force

The relationship between the applied hand force and the screw torque is

$$F_h = \frac{T_{\text{raise}}}{L_h}, \quad (278)$$

where  $L_h$  is the effective perpendicular distance between the screw axis and the point at which the operator applies the force.

For an initial effective handle length of

$$L_h = 18 \text{ in,} \quad (279)$$

the required hand force would be

$$F_h = \frac{1409}{18} \approx 78.3 \text{ lbf.} \quad (280)$$

This force was considered unnecessarily high for comfortable manual operation. Therefore, the effective handle length was increased to

$$L_h = 24 \text{ in.} \quad (281)$$

The resulting preliminary hand force is

$$F_{h,\text{raise}} = \frac{1409}{24} \approx 58.7 \text{ lbf.} \quad (282)$$

The lowering torque was previously calculated as

$$T_{\text{lower}} = 442 \text{ lbf} \cdot \text{in.} \quad (283)$$

Therefore, the corresponding lowering hand force is

$$F_{h,\text{lower}} = \frac{442}{24} \approx 18.4 \text{ lbf.} \quad (284)$$

The maximum hand force occurs at the minimum scissor angle, where the horizontal screw load reaches its maximum value. As the jack rises and the arm angle increases, the required screw force and operating torque decrease.

## 11.2 Handle-Rod Bending Check

The crank arm was preliminarily modeled as a solid circular steel rod subjected to the operating torque through an end-applied force. The maximum bending moment at the handle connection is therefore

$$M_h = F_{h,\text{raise}} L_h. \quad (285)$$

Substituting the selected handle length and operating force gives

$$M_h = (58.7)(24) \approx 1409 \text{ lbf} \cdot \text{in.} \quad (286)$$

A solid handle diameter of

$$d_h = 0.750 \text{ in} \quad (287)$$

was selected for the preliminary design.

The nominal maximum bending stress in a solid circular rod is

$$\sigma_{b,h} = \frac{32M_h}{\pi d_h^3}. \quad (288)$$

Therefore,

$$\sigma_{b,h} = \frac{32(1409)}{\pi(0.750)^3} \approx 34.0 \text{ ksi}. \quad (289)$$

The handle was assumed to be manufactured from AISI 1045 steel with a preliminary yield strength of

$$S_{y,h} = 77 \text{ ksi}. \quad (290)$$

Using a design factor of

$$n = 1.5, \quad (291)$$

the allowable bending stress is

$$\sigma_{\text{allow},h} = \frac{S_{y,h}}{n}. \quad (292)$$

Thus,

$$\sigma_{\text{allow},h} = \frac{77}{1.5} \approx 51.3 \text{ ksi}. \quad (293)$$

Since

$$\sigma_{b,h} = 34.0 \text{ ksi} < \sigma_{\text{allow},h} = 51.3 \text{ ksi}, \quad (294)$$

the selected 0.750 in solid steel handle is satisfactory with respect to nominal bending stress.

The corresponding preliminary bending design factor is

$$n_{h,\text{actual}} = \frac{S_{y,h}}{\sigma_{b,h}} = \frac{77}{34.0} \approx 2.26. \quad (295)$$

### 11.3 Handle Connection

The handle is designed as a removable component connected to the end of the power screw. The connection must transmit the full operating torque while allowing the handle to be removed for compact storage.

The preliminary connection may consist of a transverse socket or cross-hole at the screw end. The handle passes through this feature and applies torque to the screw through direct bearing contact.

The connection region should be designed so that

- the cross-hole does not intersect the highly stressed ACME thread root;
- sufficient material remains between the cross-hole and the screw end;
- the handle contact surfaces are hardened or locally reinforced;
- sharp corners and abrupt diameter transitions are avoided;
- the handle cannot unintentionally disengage during operation.

A cross-hole or socket introduces a local reduction in screw area and a geometric stress concentration. The local peak stress may be represented as

$$\sigma_{\text{max}} = K_t \sigma_{\text{nom}}, \quad (296)$$

where  $K_t$  depends on the hole diameter, screw-end diameter, edge distance, and local fillet geometry.

Accordingly, the handle-connection feature should be placed on an enlarged unthreaded screw-end section rather than directly through the 0.750 in threaded region.

## 11.4 Final Handle Selection

The preliminary handle configuration is summarized as follows:

$$L_h = 24 \text{ in}, \quad (297)$$

$$d_h = 0.750 \text{ in}, \quad (298)$$

$$F_{h,\text{raise}} \approx 58.7 \text{ lbf}, \quad (299)$$

$$F_{h,\text{lower}} \approx 18.4 \text{ lbf}. \quad (300)$$

The handle may be manufactured as a removable straight bar or as a two-piece folding crank to reduce storage length. A rotating hand grip may also be incorporated at the operator end to improve ergonomics and reduce friction between the handle and the operator's hand.

Because the calculated operating torque does not yet include axial-bearing friction, seal drag, contamination, or misalignment, the final operating force should be verified experimentally using the completed prototype.

## 12 Base Plate and Upper Saddle Checks

The lower base assembly and upper saddle transmit the vehicle load through the pivot brackets into the scissor arms. The principal dimensions used in the preliminary calculations are summarized below.

For the base assembly,

$$B_b = 2.25 \text{ in}, \quad (301)$$

$$L_b = 5.2134 \text{ in}, \quad (302)$$

$$t_b = 0.1624 \text{ in}, \quad (303)$$

$$h_b = 1.1375 \text{ in}. \quad (304)$$

For the upper saddle,

$$B_s = 1.85 \text{ in}, \quad (305)$$

$$L_s = 4.2077 \text{ in}, \quad (306)$$

$$t_s = 0.1875 \text{ in}, \quad (307)$$

$$h_s = 1.0940 \text{ in}. \quad (308)$$

The bracket-root width was measured as

$$b_r = 1.9252 \text{ in}. \quad (309)$$

The minimum operating angle of the scissor arms is

$$\theta_{\min} = 10.5^\circ, \quad (310)$$

and the maximum axial force in each arm is

$$F_{\text{arm}} = 9070 \text{ lbf}. \quad (311)$$

The load is assumed to be distributed equally between the two bracket cheeks. Therefore, the force acting on one cheek is

$$F_{\text{cheek}} = \frac{F_{\text{arm}}}{2} = 4535 \text{ lbf}. \quad (312)$$

The horizontal and vertical components acting on one cheek are

$$F_{x,\text{cheek}} = F_{\text{cheek}} \cos \theta_{\min}, \quad (313)$$

$$F_{y,\text{cheek}} = F_{\text{cheek}} \sin \theta_{\min}. \quad (314)$$

Thus,

$$F_{x,\text{cheek}} = 4535 \cos 10.5^\circ \approx 4459 \text{ lbf}, \quad (315)$$

$$F_{y,\text{cheek}} = 4535 \sin 10.5^\circ \approx 826 \text{ lbf}. \quad (316)$$

The bracket and plate material was assumed to be HSLA Grade 80 steel with

$$S_y = 80 \text{ ksi.} \quad (317)$$

Using a design factor of

$$n = 1.5, \quad (318)$$

the allowable normal stress is

$$\sigma_{\text{allow}} = \frac{80}{1.5} = 53.3 \text{ ksi.} \quad (319)$$

## 12.1 Lower Bracket Root Stress

The horizontal arm-force component produces an in-plane bending moment at the root of the lower bracket cheek:

$$M_b = F_{x,\text{cheek}} h_b. \quad (320)$$

Therefore,

$$M_b = (4459)(1.1375) \approx 5072 \text{ lbf} \cdot \text{in.} \quad (321)$$

The lower cheek is modeled as a rectangular plate bending in its own plane. The corresponding section modulus is

$$S_b = \frac{t_b b_r^2}{6}. \quad (322)$$

Thus,

$$S_b = \frac{(0.1624)(1.9252)^2}{6} \approx 0.1003 \text{ in}^3. \quad (323)$$



The nominal bending stress is

$$\sigma_{b,b} = \frac{M_b}{S_b}. \quad (324)$$

Hence,

$$\sigma_{b,b} = \frac{5072}{0.1003} \approx 50.6 \text{ ksi}. \quad (325)$$

The vertical force component produces a direct normal stress at the bracket root:

$$\sigma_{a,b} = \frac{F_{y,\text{cheek}}}{t_b b_r}. \quad (326)$$

Therefore,

$$\sigma_{a,b} = \frac{826}{(0.1624)(1.9252)} \approx 2.64 \text{ ksi}. \quad (327)$$

Assuming the direct stress acts on the tensile side of the bending stress, the maximum nominal normal stress is

$$\sigma_{\max,b} = \sigma_{b,b} + \sigma_{a,b}. \quad (328)$$

Thus,

$$\sigma_{\max,b} = 50.6 + 2.64 \approx 53.2 \text{ ksi}. \quad (329)$$

Since

$$\sigma_{\max,b} = 53.2 \text{ ksi} < \sigma_{\text{allow}} = 53.3 \text{ ksi}, \quad (330)$$

the lower bracket root marginally satisfies the nominal stress requirement.

However, the remaining margin is extremely small. This result does not include local stress concentration caused by the bracket-to-base transition, fillet geometry, forming radius, weld toe, or

local plate distortion. Consequently, the lower bracket root should be treated as a critical region and verified using the final CAD geometry and local finite-element analysis.

## 12.2 Lower Pivot Bearing Stress

The projected bearing stress in one lower bracket cheek is

$$\sigma_{br,b} = \frac{F_{cheek}}{d_p t_b}, \quad (331)$$

where

$$d_p = 0.500 \text{ in.} \quad (332)$$

Substituting the initial base thickness gives

$$\sigma_{br,b} = \frac{4535}{(0.500)(0.1624)} \approx 55.8 \text{ ksi.} \quad (333)$$

Since

$$55.8 \text{ ksi} > 53.3 \text{ ksi}, \quad (334)$$

the initial 0.1624 in thickness is insufficient at the lower pivot-hole bearing region.

The minimum required local thickness is

$$t_{req} = \frac{F_{cheek}}{d_p \sigma_{allow}}. \quad (335)$$

Therefore,

$$t_{req} = \frac{4535}{(0.500)(53.3 \times 10^3)} \approx 0.170 \text{ in.} \quad (336)$$

Accordingly, the lower pivot region was revised to provide a local effective thickness of

$$t_{b,\text{final}} = 0.1875 \text{ in.} \quad (337)$$

The revised bearing stress becomes

$$\sigma_{\text{br},b,\text{final}} = \frac{4535}{(0.500)(0.1875)} \approx 48.4 \text{ ksi.} \quad (338)$$

Therefore,

$$48.4 \text{ ksi} < 53.3 \text{ ksi}, \quad (339)$$

and the revised lower pivot region satisfies the bearing-stress requirement.

### 12.3 Upper Bracket Root Stress

The bending moment at the upper bracket root is

$$M_s = F_{x,\text{cheek}} h_s. \quad (340)$$

Thus,

$$M_s = (4459)(1.0940) \approx 4878 \text{ lbf} \cdot \text{in.} \quad (341)$$

The section modulus of the upper bracket cheek is

$$S_s = \frac{t_s b_r^2}{6}. \quad (342)$$

Therefore,

$$S_s = \frac{(0.1875)(1.9252)^2}{6} \approx 0.1158 \text{ in}^3. \quad (343)$$

The nominal bending stress is

$$\sigma_{b,s} = \frac{M_s}{S_s}. \quad (344)$$

Hence,

$$\sigma_{b,s} = \frac{4878}{0.1158} \approx 42.1 \text{ ksi}. \quad (345)$$

The direct normal stress caused by the vertical force component is

$$\sigma_{a,s} = \frac{F_{y,\text{cheek}}}{t_s b_r}. \quad (346)$$

Therefore,

$$\sigma_{a,s} = \frac{826}{(0.1875)(1.9252)} \approx 2.29 \text{ ksi}. \quad (347)$$

The maximum nominal normal stress is

$$\sigma_{\max,s} = \sigma_{b,s} + \sigma_{a,s}. \quad (348)$$

Thus,

$$\sigma_{\max,s} = 42.1 + 2.29 \approx 44.4 \text{ ksi}. \quad (349)$$

Since

$$44.4 \text{ ksi} < 53.3 \text{ ksi}, \quad (350)$$

the upper bracket root satisfies the selected nominal stress requirement.

## 12.4 Upper Pivot Bearing Stress

The projected bearing stress in one upper bracket cheek is

$$\sigma_{\text{br},s} = \frac{F_{\text{cheek}}}{d_p t_s}. \quad (351)$$

Substituting the selected upper-bracket thickness gives

$$\sigma_{\text{br},s} = \frac{4535}{(0.500)(0.1875)} \approx 48.4 \text{ ksi}. \quad (352)$$

Since

$$48.4 \text{ ksi} < 53.3 \text{ ksi}, \quad (353)$$

the upper pivot bearing region satisfies the selected design requirement.

## 12.5 Upper Saddle Contact Pressure

The effective vehicle-contact area of the upper saddle is

$$A_{\text{contact}} = 4.7250 \text{ in}^2. \quad (354)$$

The average contact pressure is

$$p_s = \frac{W}{A_{\text{contact}}}. \quad (355)$$

Thus,

$$p_s = \frac{3307}{4.7250} \approx 700 \text{ psi}. \quad (356)$$

Therefore,

$$p_s \approx 0.700 \text{ ksi}. \quad (357)$$

This average pressure is low relative to the yield strength of the steel saddle. However, the actual contact pressure may be significantly higher if the vehicle contacts only a small portion of the

nominal saddle area. Raised ribs, serrations, V-grooves, or edge contact can produce local pressure concentrations that are not represented by the average value.

## 12.6 Average Ground Pressure Under the Base

The nominal base-plate contact area is

$$A_b = B_b L_b. \quad (358)$$

Therefore,

$$A_b = (2.25)(5.2134) \approx 11.73 \text{ in}^2. \quad (359)$$

The average ground pressure is

$$p_b = \frac{W}{A_b}. \quad (360)$$

Thus,

$$p_b = \frac{3307}{11.73} \approx 282 \text{ psi}. \quad (361)$$

The average ground pressure is therefore

$$p_b \approx 0.282 \text{ ksi}. \quad (362)$$

This calculation assumes uniform support over the complete base area. Actual ground conditions may produce nonuniform pressure, edge loading, or partial support. Local base-plate bending should therefore be evaluated using the final formed geometry, flange arrangement, and realistic ground contact conditions.

## 12.7 Local Stress-Concentration Considerations

The analytical calculations presented in this section are based on nominal section properties and idealized in-plane loading. The following regions remain susceptible to local stress concentration:

- the lower bracket-to-base transition;
- the upper bracket-to-saddle transition;
- forming radii and sharp sheet-metal bends;
- pivot-hole edges;
- local reinforcement terminations;
- weld toes or heat-affected zones, if welded construction is used;
- regions subjected to eccentric or partial contact loading.

The local peak stress may be represented as

$$\sigma_{\max} = K_t \sigma_{\text{nom}}, \quad (363)$$

where  $K_t$  depends on the completed bracket geometry, bend radius, reinforcement transition, and load path.

The lower bracket root is especially critical because its nominal combined stress is already close to the allowable value. The final design should therefore incorporate smooth transitions, generous bend radii, and local reinforcement around the pivot and root regions.

A local finite-element analysis is recommended for the lower bracket root, upper saddle transition, and base-plate contact region before final design release.

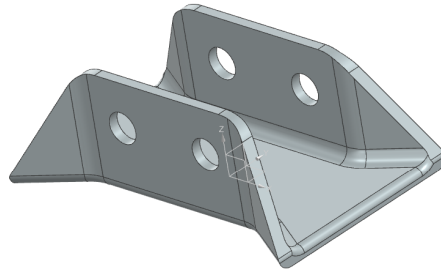


Figure 5: Bracket

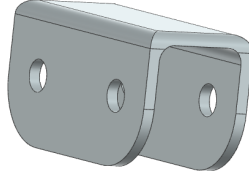


Figure 6: Saddle

## 13 Design Summary and Final Specifications

Table 2 summarizes the principal final design selections.

Table 2: Principal final design specifications

| Item                  | Final Selection                                        |
|-----------------------|--------------------------------------------------------|
| Rated vertical load   | 3307 lbf                                               |
| General design factor | 1.5                                                    |
| Arm pin-center length | 10.4857 in                                             |
| Operating-angle range | 10.5–41.1 deg                                          |
| Maximum arm force     | 9.07 kip                                               |
| Maximum screw force   | 17.8 kip                                               |
| Arm section           | $1.875 \times 0.800 \times 0.1875$ in                  |
| Arm material          | HSLA Grade 80 steel                                    |
| Main pivot pin        | 0.500 in AISI 4140 HT clevis pin                       |
| Power screw           | 3/4–6 ACME, RH, single-start                           |
| Threaded nut          | C95400 aluminum bronze, 1.75 in engagement             |
| Revised trunnion      | 0.875 in steel sleeve, 0.625 in through-pin            |
| Axial support         | 81206-type cylindrical roller thrust bearing           |
| Handle                | 24 in effective radius, 0.750 in solid AISI 1045 steel |
| Base size             | $2.25 \times 5.2134 \times 0.1624$ in                  |
| Upper saddle size     | $1.85 \times 4.2077 \times 0.1875$ in                  |

The principal design revisions resulting from the analytical checks were:

- increasing the arm-end clear ligament from 0.300 in to 0.400 in;
- adding inward-projecting support sleeves to reduce the effective bending span of the 0.500 in clevis pins;
- replacing the integral bronze trunnions with a separate steel sleeve and through-pin assembly;



- locally increasing the effective arm thickness around the trunnion sleeve to 0.200 in;
- locally increasing the lower pivot-region thickness to 0.1875 in;
- using an axial cylindrical roller bearing and positive thrust collar at the support-side block.

The most critical nominal regions are the lower bracket root, pivot-hole bearing areas, screw root, trunnion sleeve interfaces, and screw-shoulder/thrust-collar transition.

## 14 Limitations and Verification Requirements

The analytical calculations presented in this report are based on simplified mechanical models intended for preliminary design verification. The major load paths, nominal stresses, buckling behavior, bearing stresses, thread loads, and operating torque were evaluated using classical machine-design relations. However, several effects were either idealized or excluded from the analytical model.

The principal limitations of the present analysis are summarized below:

- The vehicle load was assumed to act vertically and centrally on the upper saddle.
- The scissor mechanism was assumed to remain symmetric throughout its operating range.
- Pin joints were idealized as frictionless revolute joints when determining the internal arm and screw forces.
- The arm forces were assumed to be distributed equally between paired bracket cheeks and bearing surfaces.
- Material behavior was assumed to remain linear elastic below the selected allowable-stress limits.
- Manufacturing tolerances, assembly clearances, local deformation, and residual stresses were not included explicitly.
- Friction coefficients at the ACME thread and bearing interfaces were treated as constant.
- The screw torque calculation was based primarily on thread friction and does not fully represent contamination, seal drag, misalignment, or variations in lubrication.
- Average bearing and thread-contact pressures were used. The nonuniform load distribution among the first few engaged threads was not modeled explicitly.
- The analytical pin-bending and trunnion calculations used simplified beam representations of the actual three-dimensional contact geometry.

- Local plastic deformation, edge loading, and contact indentation were not evaluated using nonlinear material or contact models.
- Fatigue was not treated as a governing design criterion because the jack is intended for low-speed, intermittent, and relatively low-cycle operation.

Several components contain geometric discontinuities that may cause local stress amplification. These include pivot-hole edges, support-sleeve ends, thread runout regions, retaining grooves, shoulder transitions, bracket roots, reinforcement terminations, and the thrust-collar connection.

The local peak stress in such regions may be expressed generally as

$$\sigma_{\max} = K_t \sigma_{\text{nom}}, \quad (364)$$

where  $K_t$  is the theoretical stress-concentration factor associated with the completed geometry.

The simplified analytical calculations do not provide a reliable value of  $K_t$  for the non-axisymmetric sheet-metal and trunnion geometries used in the jack. Therefore, the final CAD model should be used for detailed finite-element verification.

The following regions are recommended for local finite-element analysis:

- the lower bracket-to-base transition;
- the upper bracket-to-saddle transition;
- the arm-end pivot-hole and shear-out region;
- the locally reinforced trunnion-bearing region;
- the steel sleeve and bronze nut-body contact interface;
- the screw thread root and thread-runout region;
- the screw shoulder and thrust-collar transition;
- the axial-bearing seat and support-block housing;
- the handle-connection feature at the screw end.

The finite-element models should include realistic contact definitions, material properties, friction where relevant, and the final manufactured geometry. Mesh refinement should be concentrated around holes, fillets, shoulders, sleeve ends, and other load-transfer discontinuities.

In addition to numerical verification, the completed mechanism should be validated experimentally. A controlled proof-load test is recommended before the design is considered suitable for practical use.

The proof-load procedure should include the following stages:

1. operation of the unloaded jack through its complete travel;
2. verification of smooth screw rotation and self-locking behavior;
3. gradual loading to the rated load;
4. measurement of permanent deformation after unloading;
5. inspection of pins, brackets, threads, bearing seats, and reinforced regions;
6. verification of the required manual operating torque;
7. proof loading above the rated load in accordance with the selected testing standard and safety procedure.

Because lifting equipment presents a direct safety risk, the jack should not be approved for supporting a vehicle during maintenance without independent physical testing, appropriate manufacturing quality control, and compliance with the applicable lifting-equipment standards.

The present work should therefore be regarded as a preliminary engineering design and analytical verification rather than a certified production design.

## 15 Conclusion

A manually operated screw-driven scissor jack was designed for a rated vehicle mass of 1500 kg, corresponding to a vertical load of approximately 3307 lbf.

The final arm pin-center length was selected as

$$L = 10.4857 \text{ in}, \tag{365}$$

with a minimum operating angle of

$$\theta_{\min} = 10.5^\circ. \tag{366}$$

At this minimum angle, the maximum axial force in each scissor arm was calculated as

$$F_{\text{arm}} \approx 9.07 \text{ kip}, \quad (367)$$

while the maximum axial force transmitted by the power screw was

$$F_{s,\text{max}} \approx 17.8 \text{ kip}. \quad (368)$$

The scissor arms were designed as HSLA Grade 80 U-channel sections. The nominal compressive, net-section, bearing, shear-out, and global buckling checks were completed.

The initial arm-end ligament of 0.300 in was found to be insufficient against shear-out and was increased to

$$e_{\text{final}} = 0.400 \text{ in}. \quad (369)$$

The upper and lower pivot joints use 0.500 in AISI 4140 heat-treated clevis pins. Support sleeves were introduced to reduce the effective unsupported pin length and improve the pin-bending behavior.

A right-hand, single-start 3/4-6 ACME power screw was selected. The screw was shown to be self-locking under the assumed steel-bronze friction conditions.

The combined axial and torsional von Mises stress at the screw root was calculated as

$$\sigma_{\text{vm}} \approx 91.6 \text{ ksi}, \quad (370)$$

which is lower than the allowable stress of the selected heat-treated AISI 4140 steel.

The thread-only raising torque was calculated as

$$T_{\text{raise}} \approx 1409 \text{ lbf} \cdot \text{in}, \quad (371)$$

while the lowering torque was approximately

$$T_{\text{lower}} \approx 442 \text{ lbf} \cdot \text{in}. \quad (372)$$

A 24 in effective handle radius gives a preliminary maximum raising force of approximately

$$F_{h,\text{raise}} \approx 58.7 \text{ lbf.} \quad (373)$$

The initial integral bronze trunnion design was found to be unsuitable because of excessive bearing stress, cantilever bending, and stress concentration at the trunnion root.

The design was therefore revised to use a separate steel bearing sleeve and AISI 4140 through-pin. This separated the low-friction threaded function of the bronze nut from the high-load bearing and shear functions of the steel components.

An axial cylindrical roller bearing was selected to support the rotating screw at the non-threaded side block. The bearing provides axial screw location while allowing rotation under the high screw reaction force.

The lower bracket root, pivot-hole bearing regions, screw root, trunnion sleeve interfaces, and thrust-collar transition were identified as the most critical regions of the design.

Several local dimensions were revised after the analytical checks, including the arm-end ligament, lower pivot thickness, arm reinforcement, and trunnion construction.

Following these revisions, the principal nominal stress and stability checks satisfied the selected design factor of 1.5.

The completed design demonstrates the strong influence of the minimum scissor angle on the internal forces. At low arm angles, relatively modest vertical vehicle loads produce large arm and screw forces. Consequently, the lowest operating position governs the sizing of most structural and power-transmission components.

Detailed finite-element analysis of the critical regions, confirmation of the final CAD travel and clearances, and controlled physical proof-load testing are required before the jack can be considered suitable for practical or safety-critical use.

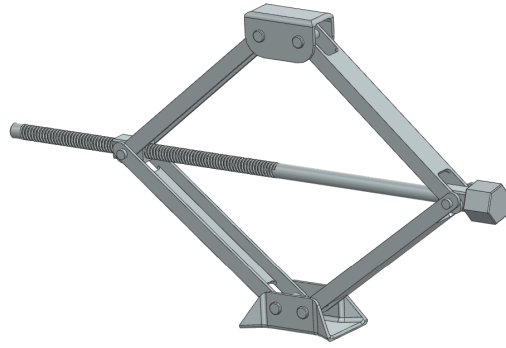


Figure 7: Car jack in the open position

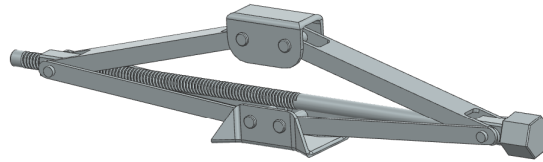


Figure 8: Car jack in the fully closed position

## 16 References

1. R. G. Budynas and J. K. Nisbett, *Shigley's Mechanical Engineering Design*, 11th ed., McGraw-Hill Education, 2020.
2. R. C. Hibbeler, *Mechanics of Materials*, Pearson.
3. T. Tevrüz, *Makina Elemanları ve Konstrüksiyon Örnekleri*, Vols. I–II.
4. E. Oberg, F. D. Jones, H. L. Horton, and H. H. Ryffel, *Machinery's Handbook*, 31st ed., Industrial Press, 2020.
5. ASME, *ASME B1.5: Acme Screw Threads*, The American Society of Mechanical Engineers.
6. ASME, *ASME B18.8.1: Clevis Pins and Cotter Pins (Inch Series)*, The American Society of Mechanical Engineers.
7. SKF, *81206 TN Single-Direction Cylindrical Roller Thrust Bearing: Product Data*, SKF Group.
8. SSAB, *ASTM A1011 HSLAS-F Grade 80: Mechanical Properties and Product Information*, SSAB Americas.

9. ASM International, *ASM Handbook, Volume 1: Properties and Selection—Irons, Steels, and High-Performance Alloys*, ASM International.
10. MatWeb, *MatWeb Material Property Data: Online Materials Database*, MatWeb, LLC.
11. Copper Development Association Inc., *C95400 Aluminum Bronze: Mechanical and Physical Properties*, Copper Development Association.
12. Siemens Digital Industries Software, *Siemens NX Documentation: Modeling, Assemblies, and Detailed Thread Features*, Siemens Digital Industries Software.
13. SKF, *Rolling Bearings Catalogue: Cylindrical Roller Thrust Bearings and Static Load Selection*, SKF Group.